C^3: Committed Confidence Calibration Selection Bias in Confidence-Based Token Commitment in Diffusion Language Models

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Abstract

Abstract to be written.

1 Introduction

2 Setup and theory

Masked diffusion decoding. A masked diffusion LM (Austin et al., 2021a; Sahoo et al., 2024; Nie et al., 2025; Ye et al., 2025) writes by unmasking. A *state* s is a sequence with a set M of masked positions. One forward pass gives, at every $i \in M$, a distribution $p_{\theta,i}$ given the observed tokens x_{obs}, with top-1 token v_i, *confidence* $c_i = p_{\theta,i}(v_i)$ and log-odds $\ell_i = \log \frac{c_i}{1-c_i}$, so $c_i = \sigma(\ell_i)$. The decoder *commits* some positions for good and repeats. The usual rule, from MaskGIT (Chang et al., 2022) and the default of LLaDA and Dream, commits the k most confident positions per step; threshold rules such as Fast-dLLM (Wu et al., 2025) commit every position above a confidence cutoff.

Committing a set $S \subseteq M$ in one step draws its tokens from the product of their marginals. With p^* the true joint of x_S given x_{obs} and p_i^* its marginals,

$$\mathrm{KL}\big(p^* \,\|\, \textstyle\prod_{i\in S} p_{\theta,i}\big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})}_{\text{(B) estimation}}, \quad (1)$$

where $\mathrm{TC}(x_S \mid x_{\text{obs}}) = \sum_{i\in S} H(p_i^*) - H(p_S^*)$ is the total correlation, zero if and only if the tokens are conditionally independent. (A) is what no product of marginals can represent; (B) is the error of the marginals themselves (cf. Ben-Hamu et al., 2025). Section 5 estimates both. C1 asks a narrower question about the marginals: once the decoder has chosen a token, is its confidence still an honest probability of being right?

The committed-token gap. Let x_0 be the true document. Under teacher forcing (§3) correctness $a_i = \mathbf{1}\{v_i = x_{0,i}\}$ is observed. A commit is a selection: the committed token had the largest ℓ among its candidates. We model a field of m candidates as a random pool $P \subseteq M$ of size m, won by $i = \arg\max_{j\in P} \ell_j$ (ties broken at random). With $\mathbb{E}_m$ the average over states, pools of size m and their winners, the *committed-token gap* is

$$\Delta(m) = \mathbb{E}_m\,[\,c - a\,], \quad (2)$$

confidence minus accuracy, so positive means overconfident.[1] $\Delta(1)$ is the ordinary signed calibration error of a random masked position; larger m stands for committing out of a larger field, as at early steps and in long outputs.

The winner's curse. The largest of m noisy estimates tends to be one whose error pushed it up: the winner's curse (Efron, 2011; Andrews et al., 2024). If it acts on decoding, $\Delta(m)$ grows with m, and a confidence cutoff means something different in exactly the states where it is applied. Whether it can act depends on what calibration conditions on. Let $\pi(\ell) = \Pr(a = 1 \mid \ell)$ be the ordinary calibration curve over all masked positions (the $m = 1$ population) and $\pi(\ell, s) = \Pr(a = 1 \mid \ell, s)$ its context-aware version.

Proposition 1 (Winning adds nothing) *Fix s. Suppose that, given s, each pair (ℓ_i, a_i) is independent of the other positions' scores $(\ell_j)_{j\neq i}$, and that the pool and the tie-breaks are drawn independently of all $(\ell_j, a_j)_{j\in M}$. Then the winner i of P satisfies*

$$\Pr\big(a_i{=}1 \;\big|\; \ell_i,\, s,\, i = \arg\max_{j\in P} \ell_j\big) = \pi(\ell_i, s).$$

Given ℓ_i and s, whether i wins depends only on the other scores, which say nothing about a_i: selection does not bias a correct posterior (Efron, 2011).

[1] The proposal used accuracy minus confidence; files from that period keep the opposite sign.

This holds however unequal the noise scales are, so heteroskedastic noise alone, the mechanism our proposal relied on, cannot make field size informative. Selection acts only through context:

Proposition 2 (Exact split of the gap) *Under the conditions of Proposition 1,*

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\textit{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\textit{field term}}, \tag{3}$$

and the field term is exactly zero at $m = 1$.

Proofs are in Appendix A.2. The first term, ordinary miscalibration reweighted towards where winners sit, can make $\Delta(m)$ grow with m with no winner's curse at all. The *field term* is nonzero only if accuracy at fixed confidence depends on context, and winning a large pool shifts which contexts winners come from (a lone high score in a weak field is more likely to be noise). It is the winner's curse in a corrected, testable form. If positions of a state share errors, Proposition 1 fails and (3) gains a third term, $\mathbb{E}_m[\pi(\ell, s) - a]$, which §3 tracks.

3 Measurement instrument

3.1 Models and data

Models. We audit LLaDA-8B (Nie et al., 2025) and Dream-7B (Ye et al., 2025), Base and Instruct, in fp16 over two RTX 2080 Ti cards, which lack bfloat16. Neither fully discloses its pretraining data, so we also trained a 110M MDLM (Sahoo et al., 2024) whose training set we control (`train110m-v3`): a bidirectional transformer with 12 layers, width 768 and 12 heads, GPT-2 BPE plus a mask token, sequence length 256 without packing, 10.0B tokens (169,055 steps) of FineWeb-Edu sample-10BT (Penedo et al., 2024). The calibration and reserve documents were removed from its training stream by ID and by hashes of their text and first 256 tokens, and no training document matches a time-split one. Its traces are computed in fp32.

Data. Base models and the 110M model read 2,000 FineWeb-Edu documents, reservoir-sampled with a fixed seed from the first 200,000 of at least 1,200 characters and truncated to their first 256 tokens under each model's tokeniser; shorter documents are left out. Instruct models read the GSM8K (Cobbe et al., 2021) and MBPP (Austin et al., 2021b) test sets with the question visible in the chat template and only the reference answer maskable; one MBPP document over 1,024 tokens is excluded. A time-split set of 500 English Wikipedia articles created after both 8B releases, unseen by every model, is a robustness check (Table 1). A document is flagged as contaminated if an 8B Base model, given its first 50 tokens, greedily reproduces a run of ≥ 32 of the next 50 (2/1,979 LLaDA, 1/1,976 Dream); 8B FineWeb endpoints are also reported without them.

Model	Text	Prec.	Docs
LLaDA-8B-Base	FineWeb-Edu	fp16	1,979
Dream-7B-Base	FineWeb-Edu	fp16	1,976
Our 110M MDLM	FineWeb-Edu	fp32	1,979
LLaDA-8B-Instruct	GSM8K	fp16	1,319
LLaDA-8B-Instruct	MBPP	fp16	499
Dream-7B-Instruct	GSM8K	fp16	1,319
Dream-7B-Instruct	MBPP	fp16	499
LLaDA-8B-Base	time-split	fp16	500
Dream-7B-Base	time-split	fp16	500
Our 110M MDLM	time-split	fp32	500
LLaDA-8B-Base	FineWeb-Edu	4-bit	198

Table 1: C1 trace runs. Top: the seven pre-registered arms. Bottom: the time-split set and the 4-bit control. In all, 0.49 billion position rows, 208 GPU-hours.

Halves. Every set is split 50/50 by document with one fixed seed (FineWeb: 1,000/1,000 before the length filter). Everything fitted (calibrators, splines, knots, temperatures) uses the fit half; every number we report comes from the report half. 500 separate reserve documents served the pilot and the non-determinism floor.

3.2 Traces

Each measurement is one forward pass on a true document with some positions masked, recording at every masked position v_i, c_i, ℓ_i, a_i, the true token's probability, entropy, margin, the top-20 and whether the position was committed. The model never fills in its own guesses, so a_i carries no knock-on errors. Each document gets three arms of states: *uniform*, each position masked independently at rate $t \in \{0.1, 0.2, \ldots, 0.9\}$, 2 masks per rate, as in training; *decoder*, every state of a teacher-forced confidence top-k trajectory, $k \in \{2, 4, 8\}$, which starts fully masked, commits the k most confident positions, fills them with the *true* tokens and repeats, so these are the states a decoder visits if it is always right; and *matched uniform*, a uniform mask of the same size as each de-

coder state, so only the pattern differs. That is 466 passes per document.

Pools are drawn from the stored trace. In every state with at least 32 masked positions we draw 20 random pools of each size $m \in \{1, 2, 4, 8, 16, 32\}$, rank them by ℓ with random tie-breaks (fp16 confidences saturate) and keep the top four ranks; $\Delta(1)$ is computed exactly over all masked positions of those states. Averages are taken within a document, every state weighing the same, then across documents.

3.3 Estimands

Calibrators. Both are fitted on the fit half. $\hat{\pi}(\ell)$ is isotonic in ℓ. $\hat{\pi}(\ell, s)$ is logistic in a restricted cubic spline of ℓ plus the state's field features (median ℓ of its masked positions, mask rate, $\log|M|$, decoding step) and their products with ℓ, so the curve's slope, not only its level, can vary with the state; the plan's additive form is a sensitivity check. Each position is weighted by $1/(|M_s| \times$ states in its document), the population the averages use, which is the $m = 1$ population of Proposition 2.

E1, E2 and the native gap. E1 is $\Delta(m)$ (2), with primary contrast $\Delta(32) - \Delta(1)$, reported beside 15-bin equal-mass ECE and smooth ECE over all masked positions. E2 is the field term $\mathbb{E}_m[\hat{\pi}(\ell) - \hat{\pi}(\ell, s)]$, with primary contrast field(32)$-$field(1). On the same winners and resamples, $\Delta = \mathbb{E}_m[c - \hat{\pi}(\ell)] + \text{field} + \mathbb{E}_m[\hat{\pi}(\ell, s) - a]$ holds exactly; the last term, zero under Proposition 1, is a specification check, beside the intra-class correlation of residuals $a - \hat{\pi}(\ell, s)$ within states. The *native* gap is $c - a$ over tokens the decoder actually committed, grouped by $|M_t|$ at commit time.

Eq. 2/3 diagnostics. On pool winners (Eq. 2) and the top four ranks (Eq. 3),

$$\Pr(a_i{=}1) = \sigma\big(\alpha\,[{+}\alpha_s] + f(\ell_i) + \beta_z z_{r,m}\big), \quad (4)$$

$$\Pr(a_i{=}1) = \sigma\big(\alpha\,[{+}\alpha_s] + f(\ell_i) + \textstyle\sum_{k=2}^{4} \gamma_k \mathbf{1}\{r{=}k\}\big), \quad (5)$$

with $z_{r,m}$ the expected r-th largest of m standard normals and f a restricted cubic spline with 9 knots (5 and 13 as sensitivity), fitted on the fit half and carried over as a fixed offset. Each is fitted without and with state fixed effects α_s, which absorb the field term because it lives between states: $\beta_z < 0$ without them and ≈ 0 with them points to a field effect, $\beta_z \neq 0$ with them to shared errors. Simulation (§3.5) shows this reading is approximate, so β_z is a diagnostic.

E3, trajectory shift. For each decoder arm we fit, on its positions and a reference arm's, $\Pr(a{=}1) = \sigma(\alpha_b + f(\ell) + \mathit{dec} \cdot (g_0 + f'(\ell)))$, with α_b an intercept per mask-rate bin (centred on $0.1, \ldots, 0.9$). The reference is the matched uniform arm (secondary: the uniform grid, reweighted to the decoder's rate mix). The endpoint is $\mathbb{E}_{\text{dec}}[\pi_{\text{ref}}(\ell, b) - \pi_{\text{dec}}(\ell, b)]$, positive if decoder-state tokens are right less often at equal confidence and mask rate; (g_0, f') are tested jointly by a sup-t test on the difference curve.

T8 controls. Each control is a row giving the E1 and E2 contrasts without and with it, and their paired difference. *Position* (absolute position, distance to the nearest unmasked token, PC-Sampler's position weight; Huang et al., 2025) and *token type* (the true token is punctuation, a stop-word or content) are covariates x: $\mathbb{E}_m[\pi(\ell) - \pi(\ell, x)]$ is removed from $\Delta(m)$ and x enters both field-term calibrators. *Temperature* fits $c = \sigma(\ell/T)$ per mask-rate bin on the fit half (a softmax temperature is a sensitivity row). *Precision* compares 200 LLaDA-8B-Base documents in 4-bit NF4 and fp16.

3.4 Inference

Bootstrap. The document is the unit. Intervals are 95% percentile intervals over 10,000 document resamples, one matrix per run shared by all pool sizes, estimands and arms, so contrasts get intervals by differencing draws. Eq. 2/3 refit per resample, so they use 1,000 (knot panels: 200); 10,000 would take weeks per run. Calibrators are not refitted per resample, so E2 intervals are conditional on the fit half.

Declaring an effect. An effect is declared only if its interval excludes 0 *and* its magnitude exceeds twice the non-determinism floor, the largest change in ℓ and $\Delta(m)$ over five repeats of 20 reserve documents per model family. Repeats were bitwise identical, so every floor is 0 and the floor half of the rule is vacuous: in practice, the interval excludes 0. $\delta_{\min} = 0.02$ was fixed in advance for power only.

Multiple testing. Holm's correction, on two-sided bootstrap p-values, runs in three families: the E1 contrast and the E2 contrast, each over the

seven arms of Table 1 in the uniform and k=4 decoder states (14 tests each), and $\gamma_2, \gamma_3, \gamma_4$ within each arm. $k \in \{2, 8\}$ and matched uniform are exploratory; the contamination-clean and time-split runs are robustness checks outside the families.

3.5 Validation and pre-registration

Known truth (T0). Before any model data, the whole pipeline ran on simulated traces in four worlds: calibrated everywhere; miscalibrated the same way in every context; miscalibrated depending on context (the winner's curse of Proposition 2); and errors shared within a state (violating Proposition 1). The E1 contrast was zero only in the first, and β_z without fixed effects strongly negative only in the third. Three departures from the plan shape how we read §4: the field-term estimator reads high, slightly positive when the truth is zero and above it otherwise; fixed effects do not fully clean β_z in the third world; and shared errors give a *negative* field term with a *positive* β_z without fixed effects. A spline with too few knots let curvature load onto $z_{r,m}$, so the knots were fixed at 9 before any model data.

Pre-registration. All choices above and the exact commands were frozen at git tag `prereg-v1` before any report-half number was computed. `prereg/deviations.md` lists departures from the plan: here, fp16, top-k-only E3, 1,000 Eq. 2/3 resamples, and a contamination threshold (the plan's suggested value) coded after the screens ran. An exploratory notebook read report-half rows of the 110M runs before the tag; its numbers were not committed or shared, and no frozen choice changed after it.

4 Selection and confidence (C1)

5 Two sources of error (C2)

Committing a set S of masked tokens in one step samples them from the product of their marginals. Given the context x_{obs}, the KL from the true joint to that product splits exactly (Eq. 1, §2) into a dependency term (A) $= \text{TC}(x_S \mid x_{\text{obs}}) = \sum_{i \in S} H(p_i^*) - H(p_S^*)$, which no product of marginals can represent, and an estimation term (B) $= \sum_{i \in S} \text{KL}(p_i^* \,\|\, p_{\theta,i})$, the error of the marginals themselves (cf. Ben-Hamu et al., 2025). Work on parallel decoding targets (A) and takes the marginals as accurate (Kim et al., 2026; Ringel et al., 2026; Azangulov et al., 2026; Kang et al., 2026). We compute both exactly on a probe with known p^*, then test real-text proxies against it.

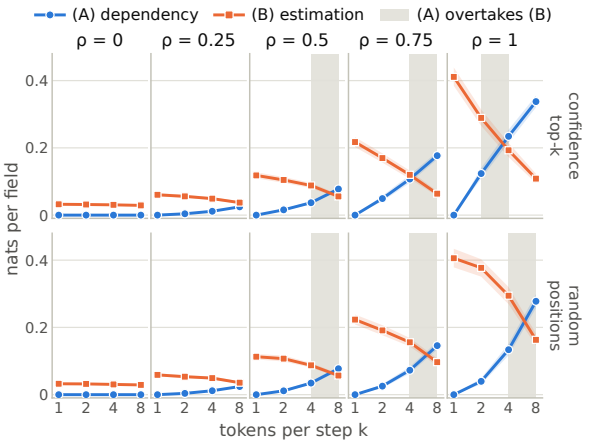


Figure 1: (A) and (B) per field on the probe, with 95% intervals, against tokens per step k (main model). Columns: ρ; rows: selection rule.

5.1 A JSON probe with known p^*

The probe fills the simple fields of 200 SchemaStore schemas (SchemaStore contributors, 2026), 6–16 per schema, with single-token values. Each field has a seeded Dirichlet(1) marginal, and fields are paired: with probability ρ, stated in a marker at the start of the instance, the follower is a fixed function of its leader's value, otherwise an independent draw, so p^* is known exactly. We fine-tune our 110M MDLM (§3) on 200k instances and evaluate 1,000 (200 per $\rho \in \{0, .25, .5, .75, 1\}$); the main model is the fine-tune that never saw the marker masked (Appendix A.1). Decoding starts with every value masked. At each step the rule picks $k \in \{1, 2, 4, 8\}$ fields, the most confident (top-1 probability) or at random; we compute (A) and (B) for that set, with $p_{\theta,i}$ the model's probability of each allowed value (mass off the list counts against (B)), and fill it with the true values (teacher forcing). Terms are summed over steps, divided by the number of fields and bootstrapped over instances (10,000 resamples).

5.2 (A) overtakes (B) at high ρ and k

(A) is zero at $k = 1$ and at $\rho = 0$ and grows with both (Figure 1). For $\rho \geq 0.5$ it overtakes (B) at $k = 8$, or at $\rho = 1$ under confidence top-k already at $k = 4$; each crossover is also where the 95% interval of (A)−(B) clears zero. For $\rho \leq 0.25$ (B) dominates up to $k = 8$. At $\rho = 1$, $k = 8$ (confidence), (A) is 0.338 [0.320, 0.355] nats per field

and (B) 0.109 [0.095, 0.123]: 76% of the total.

Confidence selection raises (A), not (B). The proposal expected ranking on noisy confidences to inflate (B), but where (B) is higher than at random the intervals overlap. (A) rises instead ($\rho = 1$, $k = 2$: 0.124 [0.108, 0.140] against 0.040 [0.031, 0.049]): the ranking commits leaders together with their followers more often than chance, which brings the crossover forward.

The terms trade off. ρ was meant to move only (A), but (B) rises too: at $k = 1$, from 0.032 at $\rho = 0$ to 0.411 at $\rho = 1$. The model reads the marker only in part (Appendix A.1), so a follower decoded after its leader misses the sharp marginal the copy implies; decoded with its leader, it has a broad true marginal, which the model matches, and the gap appears as (A) instead. At $\rho = 1$, from $k = 1$ to 8, (B) falls from 0.411 to 0.109 while (A) rises from 0 to 0.338. Without fine-tuning, (B) dominates everywhere ((A) is $\leq 4\%$ of the total).

5.3 Real text: proxies checked on the probe

These exact values hold for one synthetic distribution and one 110M model, not for real text, where p^* is unknown and we use proxies. For (A), parallel–sequential disagreement: the fraction of the k most confident masked positions whose token changes if they are committed one at a time (earlier ranks filled with the model's own tokens) instead of in one pass. For (B), the gap $\Delta = c - a$ (positive means overconfident) and the cross-entropy (CE) of the true token, at the same positions. We run the 110M model (fp32) and LLaDA-8B-Base (fp16; Nie et al., 2025) on 300 fit-half FineWeb-Edu documents (Penedo et al., 2024), one state at each mask rate $0.1, 0.2, \ldots, 1$ (Appendix A.1), and the same proxy on the probe.

The proxy tracks (A) weakly. Over the 15 confidence-rule cells with $k \geq 2$, its Spearman correlation with the exact (A) is 0.49 [0.28, 0.63] for the main model and 0.28 [0.06, 0.49] for the other. It rises with k, barely with ρ: at $k = 8$, 0.028 at $\rho = 0$, where (A) is zero, and 0.049 at $\rho = 1$.

On real text the (A) proxy is near zero. Over mask rates 0.1–0.9,[2] disagreement at $k = 8$ is 0.0021 for the 110M model and 0.0022 for LLaDA, and at most 0.0003 at $k \leq 4$. CE grows from 0.08 nats at $k = 1$ to 0.40 at $k = 8$ (LLaDA: 0.08 to 0.23), though it includes the text's own entropy and lower ranks are harder positions. Δ cannot be told from zero at $k = 1$ and is small but positive at $k = 8$: $+0.007$ [0.004, 0.010] and, for LLaDA, $+0.004$ [0.002, 0.007].

So at 2–8 tokens per step real text looks like the probe's low-ρ regime, where (B) dominates. Two limits: the proxy sees only dependence the model has learned, so near-zero disagreement does not show that (A) is small, and the proxies' units differ: real text gives a direction, not a split in nats.

6 C3

7 Related work

Masked diffusion LMs. LLaDA and Dream (Nie et al., 2025; Ye et al., 2025) scale masked diffusion (Sahoo et al., 2024) to 7–8B parameters and commit the most confident positions, as MaskGIT did (Chang et al., 2022). We audit that rule; we propose no model.

Parallel and confidence-based decoders. Decoders replace top-1 probability, which can misstate uncertainty (Kim et al., 2025a), with entropy, margin or stability (Ben-Hamu et al., 2025; Kim et al., 2025a,b), tune the cutoff (Wu et al., 2025; Shen and Ro, 2025) or learn the policy (Jazbec et al., 2026). They evaluate speed and accuracy, not the calibration of the tokens they commit, which C1 measures; C3a thresholds a recalibrated probability instead (§6).

Calibration and the winner's curse. Calibration is usually measured over all predictions (Guo et al., 2017). The closest work, PC-Sampler (Huang et al., 2025), recalibrates the commit statistic for a positional bias; we add its weight as a covariate. Keeping the largest of noisy estimates biases it upward (Andrews et al., 2024) but leaves a correct posterior unbiased (Efron, 2011), which confines our winner's curse to the field term (§2).

Dependency between tokens. Dependency-aware decoders (Kim et al., 2026; Ringel et al., 2026; Azangulov et al., 2026) and ParallelBench (Kang et al., 2026) target term (A) of Eq. 1 and take the marginals as accurate. The split is not ours: EB-Sampler uses the same two terms (Ben-Hamu et al., 2025), and Tang et al. (2026) note that sampler and model error stay entangled.

[2] At rate 1 every input is the same all-`[MASK]` sequence. That one state gives the 110M model nearly all of its disagreement over all ten rates (0.0644 at $k = 8$); LLaDA does not disagree there.

C2 computes both in nats where p^* is known (§5).

Limitations

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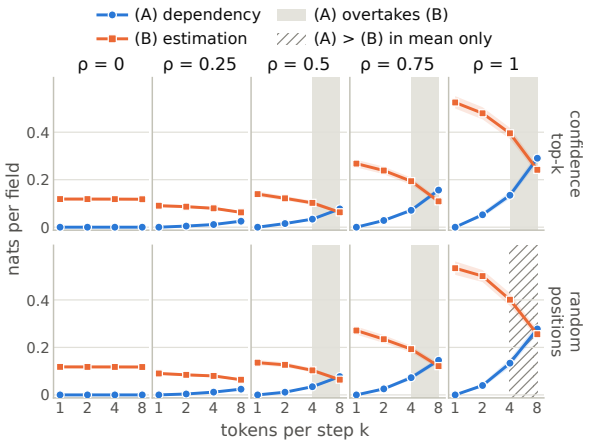


Figure 2: As Figure 1, for the fine-tune that saw the marker masked. Hatched: (A) > (B) in mean only.

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A Supplementary details

A.1 C2: the main model and T10 details

Main model. One fine-tune masked the ρ marker like any token (100M tokens); the main model never did, as at evaluation (300M). Neither moved the copied value's probability by our bar of 0.5 when the marker is switched from $\rho = 1$ to 0 with only the leader observed (0.014 and 0.251), so we took the lower validation NELBO (0.1751 against 0.1892 nats per token); the bar was first written down with these results, the fallback only when applied. The other (Figure 2) crosses over at $k = 8$ for every $\rho \geq 0.5$ under both rules.

T10. Unlike the plan, we skip documents under 256 tokens (fit-half ids up to 601; 600 for LLaDA) and keep at least 8 masked positions per state. Intervals resample documents (10,000 times).

A.2 Proofs of Propositions 1 and 2

Setting. Let s be a state with masked set M, $\ell_{-i} = (\ell_j)_{j \in M \setminus \{i\}}$ and $\pi(\ell_i, s) = \Pr(a_i{=}1 \mid \ell_i, s)$ (it may also depend on i; we suppress the index). A pool $P \subseteq M$ is drawn independently of $(\ell_j, a_j)_{j \in M}$ given s, ties are broken by independent randomness U, and W_i is the event that $i \in P$ wins, $i = \arg\max_{j \in P} \ell_j$.

Proposition 1. Suppose that, given s, (ℓ_i, a_i) is independent of ℓ_{-i}. Then $\Pr(a_i{=}1 \mid \ell_i, s, W_i) = \pi(\ell_i, s)$ almost surely on W_i.

Proof. Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$. Given s, (P, U) is independent of $(\ell_j, a_j)_{j \in M}$ and (ℓ_i, a_i) of ℓ_{-i}, so the joint law of $(\ell_i, a_i, \ell_{-i}, P, U)$ factorises as $p(P, U \mid s)\, p(\ell_i, a_i \mid s)\, p(\ell_{-i} \mid s)$: $(\ell_i, a_i) \perp Y \mid s$. By weak union $a_i \perp Y \mid Z$, that is, $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$ a.s. The event W_i is a function of (Y, Z), so for every bounded measurable h,

$$\begin{aligned} \mathbb{E}[a_i\, h(Z)\, \mathbf{1}_{W_i}] &= \mathbb{E}\big[\mathbb{E}[a_i \mid Y, Z]\, h(Z)\, \mathbf{1}_{W_i}\big] \\ &= \mathbb{E}[\pi(\ell_i, s)\, h(Z)\, \mathbf{1}_{W_i}]. \end{aligned} \tag{6}$$

As h ranges over bounded functions of Z, this says $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s. on W_i. Nothing requires the pairs to be identically distributed, so the result holds for any noise scales. □

Proposition 2. Let $\mathbb{E}_m$ average over states (weighted by any function of s), pools of size m and tie-breaks, and let (ℓ, a) be the winner's pair, so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - a]$. Let $\pi(\ell) = \Pr_1(a{=}1 \mid \ell)$ be the calibration curve of the $m = 1$ population, where the winner is a uniformly random masked position. Under the hypothesis of Proposition 1, $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell)] + \mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]$, and the second term is zero at $m = 1$.

Proof. Exactly one W_i holds in each pool. Applying (6) with $h \equiv 1$, summing over $i \in M$ and averaging over states,

$$\begin{aligned} \mathbb{E}_m[a] = \textstyle\sum_i \mathbb{E}_m[a_i \mathbf{1}_{W_i}] &= \textstyle\sum_i \mathbb{E}_m[\pi(\ell_i, s) \mathbf{1}_{W_i}] \\ &= \mathbb{E}_m[\pi(\ell, s)], \end{aligned}$$

so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$; adding and subtracting $\pi(\ell)$ gives the split. At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property, since π is the calibration curve of exactly this population, and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ by the display above, so the field term is zero. □

The estimator inherits the last step only if $\hat{\pi}(\ell)$ is fitted on the $m = 1$ population with the same state weights the averages use, which is why both calibrators are fitted with pool weights (§3.3).